

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 2: Growth vs Development and Inside the Seed: Maize H614 and Rosecoco Beans

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	A	B	A	C	B

USE AFTER: PREDICT PHASE — Predict: classifying the 6 changes from Mr. Wafula's field as G, D or Both (change 3, unifoliate to trifoliate leaves)

Q1. On Mr. Wafula's farm, a bean plant makes its first single-bladed (unifoliate) leaves on Day 12 and three-part (trifoliate) leaves on Day 18. How should this change be classified?

- A. Growth only, because the plant has become bigger
- B. Development, because the leaf changes in form and becomes more complex through differentiation**
- C. Growth only, because there are now more leaves
- D. Neither growth nor development, because leaves are not part of the seed

Correct answer: B. A change from unifoliate to trifoliate leaves is a change in form and complexity through cell differentiation, which is development. It is not simply an increase in size.

USE AFTER: PREDICT PHASE — Predict: change 5, a maize seed swelling after the 25 mm rain (Think-Pair-Share, 'Does swelling of a seed in water count as growth?')

Q2. A dry maize grain swells after 25 mm of rain falls on the field. Why is this swelling NOT counted as true growth?

- A. Swelling makes new organs form, so it is development, not growth
- B. Growth only means getting taller, and the grain did not get taller
- C. Maize grains are not living things, so they cannot grow
- D. Water can be lost again, so the increase is not a permanent, irreversible increase in size or dry mass**

Correct answer: D. Growth is a permanent, irreversible increase in size or dry mass. A seed swollen with water can dry out and shrink again, so the swelling is not true growth.

USE AFTER: OBSERVE PHASE — Observe step 5: using the Day 14 to Day 42 stem width data as growth examples and calculating the increase in stem width

Q3. The stem of a maize plant is 6 mm wide on Day 14 and 22 mm wide on Day 42. By how much did the stem width increase, and is this growth?

- A. 16 mm; it is growth, because the size increased permanently**
- B. 28 mm; it is growth, because the stem became thicker
- C. 16 mm; it is not growth, because the plant did not get taller
- D. 22 mm; it is development, because the stem changed form

Correct answer: A. Increase = $22 - 6 = 16$ mm. A permanent increase in size is growth, even when it is thickening and not height.

USE AFTER: OBSERVE PHASE — Observe step 2 and 4: splitting the soaked bean into two cotyledons and completing the Part I Where found | Function table

Q4. A learner splits a soaked Rosecoco bean seed into two halves. Which part of the bean stores the food that the young plant will use?

- A. The testa
- B. The two cotyledons**
- C. The micropyle
- D. The hilum

Correct answer: B. In a dicot such as the bean, the two large cotyledons store the food. The testa protects the seed, the micropyle lets water in, and the hilum is the scar where the seed was attached.

USE AFTER: OBSERVE PHASE — Observe step 2: cutting the maize H614 grain lengthwise to find the coleoptile, coleorhiza, scutellum and endosperm

Q5. In a maize grain, which part protects the shoot tip (plumule) as it pushes up through the soil?

- A. The coleoptile**
- B. The coleorhiza
- C. The scutellum
- D. The endosperm

Correct answer: A. The coleoptile covers the plumule and protects it as it pushes through the soil. The coleorhiza protects the radicle, the scutellum absorbs food, and the endosperm stores food.

USE AFTER: EXPLAIN PHASE — Explain: using seed structure to explain how the bean (cotyledons) and maize (endosperm and scutellum) store and use food

Q6. Which statement correctly describes how a maize grain (a monocot) gets food for its embryo?

- A. Food is stored in two large cotyledons, as in a bean
- B. The endosperm is the embryo, and it is the part that grows into the plant
- C. Food is stored in the endosperm, and the single cotyledon (scutellum) absorbs it for the embryo**
- D. Maize has no cotyledon, so the embryo makes all its own food

Correct answer: C. Maize stores food in the endosperm. The one cotyledon, the scutellum, absorbs this food and passes it to the embryo. The endosperm is a food store and not the embryo.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 drawing with arrows showing radicle becoming root and plumule becoming shoot

Q7. A learner revises their Day 1 model by adding labelled seed drawings and arrows inside the soil. Which arrows show the correct development of the embryo parts?

- A. Radicle becomes the leaf; plumule becomes the root
- B. Radicle becomes the root; plumule becomes the shoot**
- C. Radicle becomes the shoot; plumule becomes the root
- D. Radicle becomes the testa; plumule becomes the cotyledon

Correct answer: B. The radicle develops into the root system and the plumule develops into the shoot with leaves.